

Chemical Bonding & Molecular Structure

A. Ionic (Electrovalent) Bonding

1. According to Fajan's rules, covalent character in an otherwise ionic compound is favoured by:
(A) a large cation and a large anion
(B) a small, highly charged cation and a large anion
(C) a large cation and a small anion
(D) a small cation and a small anion
2. The lattice enthalpy of an ionic compound is defined as the energy required to:
(A) form one mole of the ionic compound from its constituent gaseous ions
(B) separate one mole of the solid ionic compound completely into its gaseous ions
(C) melt one mole of the ionic solid
(D) dissolve one mole of the ionic solid in water
3. Which of the following ionic compounds has the highest lattice enthalpy?
(A) NaCl
(B) MgO
(C) KCl
(D) CaCl₂
4. According to Fajan's rules, a cation with a pseudo-noble-gas (18-electron) configuration, such as Cu⁺, tends to cause:
(A) greater ionic character than an inert-gas-configuration cation of similar size
(B) greater covalent character, owing to its higher polarising power
(C) no change in the covalent character of the bond
(D) the accompanying anion to become smaller
5. Ionic bonds are generally formed between elements that have:
(A) nearly identical electronegativities
(B) a large difference in electronegativity, typically a metal and a non-metal
(C) two non-metals of similar atomic size
(D) two metals of similar reactivity

B. Covalent Bonding, Lewis Structures & Formal Charge

6. According to the octet rule, atoms tend to gain, lose or share electrons so as to attain:
(A) a duplet configuration only
(B) eight electrons in their valence shell, resembling the nearest noble gas
(C) a completely filled d-subshell
(D) an overall zero formal charge
7. Which of the following molecules does not obey the octet rule, having an electron-deficient central atom?
(A) CH₄
(B) BF₃
(C) NH₃
(D) H₂O

8. Which of the following species has an expanded octet on its central atom?
- (A) CO_2
 - (B) SF_6
 - (C) BeCl_2
 - (D) NH_3
9. The formal charge on an atom in a Lewis structure is calculated using the expression:
- (A) $Q = (\text{total valence electrons}) - (\text{bonding electrons})$
 - (B) $Q = (\text{valence electrons of the free atom}) - (\text{non-bonding electrons}) - \frac{1}{2}(\text{bonding electrons})$
 - (C) $Q = (\text{bonding electrons}) - (\text{non-bonding electrons})$
 - (D) $Q = (\text{number of bonds}) - (\text{number of lone pairs})$
10. In the Lewis (resonance) structure of the carbonate ion, CO_3^{2-} , the average formal charge on each of the three oxygen atoms is:
- (A) 0
 - (B) -1
 - (C) $-2/3$
 - (D) $-1/3$

C. VSEPR Theory & Molecular Shapes

11. According to VSEPR theory, a central atom with four bond pairs and no lone pairs adopts a:
- (A) square planar geometry
 - (B) tetrahedral geometry
 - (C) trigonal bipyramidal geometry
 - (D) octahedral geometry
12. SF_4 has four bond pairs and one lone pair on the central sulfur atom; its molecular shape is:
- (A) tetrahedral
 - (B) trigonal bipyramidal
 - (C) see-saw
 - (D) square pyramidal
13. ClF_3 , with three bond pairs and two lone pairs on chlorine, has a molecular shape best described as:
- (A) trigonal planar
 - (B) T-shaped
 - (C) pyramidal
 - (D) bent
14. XeF_4 , with four bond pairs and two lone pairs on xenon, has a molecular geometry of:
- (A) octahedral
 - (B) square planar
 - (C) tetrahedral
 - (D) square pyramidal
15. Which of the following molecules is T-shaped?
- (A) BF_3
 - (B) ClF_3
 - (C) SF_4
 - (D) XeF_4

16. The H–O–H bond angle in water is smaller than the ideal tetrahedral angle of 109.5° mainly because:
- (A) oxygen is more electronegative than hydrogen
 - (B) lone pair–lone pair repulsion exceeds bond pair–bond pair repulsion, compressing the bond angle
 - (C) hydrogen bonds pull the atoms apart
 - (D) water is a polar molecule
17. According to VSEPR theory, the correct order of repulsive strength between electron pairs is:
- (A) bond pair–bond pair > lone pair–bond pair > lone pair–lone pair
 - (B) lone pair–lone pair > lone pair–bond pair > bond pair–bond pair
 - (C) lone pair–bond pair > lone pair–lone pair > bond pair–bond pair
 - (D) all three types of repulsion are equal
18. IF_5 , which has five bond pairs and one lone pair on iodine, has a molecular shape of:
- (A) trigonal bipyramidal
 - (B) octahedral
 - (C) square pyramidal
 - (D) pentagonal planar

D. Hybridisation

19. The hybridisation of the carbon atom in methane, CH_4 , is:
- (A) sp
 - (B) sp^2
 - (C) sp^3
 - (D) sp^3d
20. The hybridisation of beryllium in BeCl_2 is:
- (A) sp
 - (B) sp^2
 - (C) sp^3
 - (D) dsp^2
21. The hybridisation of sulfur in SF_6 is:
- (A) sp^3
 - (B) sp^3d
 - (C) sp^3d^2
 - (D) dsp^2
22. Which of the following pairs of species have the same hybridisation on their central atom?
- (A) BF_3 and NH_3
 - (B) CO_2 and BeCl_2
 - (C) CH_4 and PCl_5
 - (D) H_2O and CO_2
23. The electronegativity of a carbon atom is highest when the carbon is in which hybridisation state?
- (A) sp^3
 - (B) sp^2
 - (C) sp
 - (D) all three states have equal electronegativity

24. The central phosphorus atom in PCl_5 is hybridised as:

- (A) sp^3
- (B) sp^3d
- (C) sp^3d^2
- (D) sp^2

E. Dipole Moment & Bond Polarity

25. Which of the following molecules has a net dipole moment of zero despite containing polar bonds?

- (A) NH_3
- (B) H_2O
- (C) CO_2
- (D) HCl

26. The dipole moment of NF_3 is lower than that of NH_3 mainly because:

- (A) fluorine is less electronegative than hydrogen
- (B) in NF_3 the bond dipoles point in a direction that partly opposes the lone-pair dipole, unlike in NH_3
- (C) nitrogen has no lone pair in NF_3
- (D) NF_3 has a larger bond angle than NH_3

27. The percentage ionic character of a bond can be estimated from the ratio of:

- (A) the observed dipole moment to the dipole moment calculated assuming 100% ionic character
- (B) bond length to bond order
- (C) electronegativity to bond enthalpy
- (D) bond angle to dipole moment

28. Among CO_2 , SO_2 and H_2O , the molecule(s) that possess a non-zero dipole moment are:

- (A) only CO_2
- (B) only SO_2 and H_2O
- (C) only H_2O
- (D) all three

29. Which of the following pairs consists of two molecules that both possess a permanent dipole moment?

- (A) CO_2 and BF_3
- (B) NH_3 and H_2O
- (C) CH_4 and CO_2
- (D) BeCl_2 and CO_2

30. Two AB_3 -type molecules differ in that one is trigonal planar with zero dipole moment while the other is pyramidal with a non-zero dipole moment. This difference is mainly due to:

- (A) different bond lengths in the two molecules
- (B) the presence or absence of a lone pair on the central atom
- (C) different atomic masses of A in the two molecules
- (D) different bond angles, unrelated to any lone pair

F. Valence Bond Theory, Resonance & Bond Parameters

31. Resonance structures of a molecule or ion differ only in the:
- (A) positions of the atoms
 - (B) arrangement of electrons (bonding pairs and lone pairs), while atomic positions stay fixed
 - (C) total number of atoms
 - (D) overall molecular formula
32. The actual structure of a resonance hybrid is best described as:
- (A) identical to any single one of the contributing resonance structures
 - (B) a weighted average of all contributing structures, generally more stable than any one of them
 - (C) less stable than any single contributing structure
 - (D) non-existent in reality
33. Bond order is often described as an inverse measure of:
- (A) bond length
 - (B) bond enthalpy
 - (C) the number of bonds between two atoms
 - (D) molecular mass
34. Among a $\text{C}\equiv\text{C}$, a $\text{C}=\text{C}$ and a $\text{C}-\text{C}$ bond, the correct order of increasing bond length is:
- (A) $\text{C}\equiv\text{C} < \text{C}=\text{C} < \text{C}-\text{C}$
 - (B) $\text{C}-\text{C} < \text{C}=\text{C} < \text{C}\equiv\text{C}$
 - (C) $\text{C}=\text{C} < \text{C}-\text{C} < \text{C}\equiv\text{C}$
 - (D) $\text{C}\equiv\text{C} < \text{C}-\text{C} < \text{C}=\text{C}$
35. As bond order between two atoms increases, bond enthalpy generally:
- (A) decreases
 - (B) increases
 - (C) remains unchanged
 - (D) becomes negative
36. Which of the following is not a general property of covalent compounds?
- (A) comparatively low melting and boiling points relative to ionic compounds
 - (B) generally poor electrical conductivity in both the solid and molten states
 - (C) high solubility in polar solvents such as water, for all covalent compounds without exception
 - (D) directional nature of the bonds formed

G. Molecular Orbital Theory

37. According to Molecular Orbital Theory, the bond order of a diatomic species is calculated as:
- (A) $N_b - N_a$
 - (B) $(N_b - N_a)/2$
 - (C) $(N_b + N_a)/2$
 - (D) N_b/N_a
38. According to MOT, the bond order of the N_2 molecule is:
- (A) 1
 - (B) 2
 - (C) 3
 - (D) 2.5

39. The O_2 molecule is paramagnetic because, according to MOT, it has:
- (A) all of its electrons paired
 - (B) two unpaired electrons in its degenerate π^* antibonding orbitals
 - (C) no electrons occupying antibonding orbitals
 - (D) a bond order of 3
40. The bond order of a hypothetical He_2 molecule, as predicted by MOT, is:
- (A) 0
 - (B) 1
 - (C) 0.5
 - (D) 2
41. Removing one electron from O_2 to form O_2^+ causes the bond order to:
- (A) decrease from 2 to 1.5
 - (B) increase from 2 to 2.5
 - (C) remain unchanged at 2
 - (D) decrease to 1
42. Which of the following diatomic species has a bond order of exactly 1?
- (A) N_2
 - (B) O_2
 - (C) Li_2
 - (D) He_2
43. Adding one electron to N_2 to form N_2^- would cause the bond order to:
- (A) increase
 - (B) decrease
 - (C) remain the same
 - (D) become zero
44. Among the following, the species with a bond order of 0.5 is:
- (A) H_2
 - (B) H_2^+
 - (C) He_2
 - (D) O_2

H. Hydrogen Bonding

45. Hydrogen bonding is most likely to occur between a hydrogen atom bonded to a highly electronegative atom and a lone pair on:
- (A) any carbon atom
 - (B) a highly electronegative atom such as F, O or N on a neighbouring molecule
 - (C) a metal cation
 - (D) a noble gas atom
46. Intramolecular hydrogen bonding (within a single molecule) is observed in:
- (A) ortho-nitrophenol
 - (B) para-nitrophenol
 - (C) sodium chloride
 - (D) carbon dioxide

47. The unusually high boiling point of water compared with the other Group 16 hydrides (H_2S , H_2Se , H_2Te) is mainly due to:

- (A) its low molar mass
- (B) extensive intermolecular hydrogen bonding between water molecules
- (C) its high dipole moment considered in isolation
- (D) covalent bonding between separate water molecules

I. Diborane & Electron-Deficient Compounds

48. Diborane, B_2H_6 , is best described as an example of:

- (A) an electron-precise hydride
- (B) an electron-deficient hydride, containing two three-centre two-electron bonds
- (C) an electron-rich hydride
- (D) an ionic hydride

49. In the structure of diborane, the bridging B–H–B bonds are commonly known as:

- (A) sigma bonds
- (B) pi bonds
- (C) banana bonds (three-centre two-electron bonds)
- (D) coordinate bonds

J. Miscellaneous

50. Isoelectronic species are defined as species that have:

- (A) the same number of atoms
- (B) the same number of electrons, and often comparable structures
- (C) the same number of protons
- (D) identical molecular formulas

Answer Key

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	11	B	21	C	31	B	41	B
2	B	12	C	22	B	32	B	42	C
3	B	13	B	23	C	33	A	43	B
4	B	14	B	24	B	34	A	44	B
5	B	15	B	25	C	35	B	45	B
6	B	16	B	26	B	36	C	46	A
7	B	17	B	27	A	37	B	47	B
8	B	18	C	28	B	38	C	48	B
9	B	19	C	29	B	39	B	49	C
10	C	20	A	30	B	40	A	50	B